

Conceptual replication study and meta-analysis suggest simulated nature does not reliably restore pure executive attention measured by the attention network task

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ABSTRACT

The ability to replicate research findings is a cornerstone of science. Recent large-scale attempts to reproduce published psychology studies have resulted in replication of fewer than half of the statistically significant results (Open Science Collaboration, 2015). In this article, we present evidence of a failure to conceptually replicate a study published in *Psychological Science* (Berman, Jonides, & Kaplan, 2008, Experiment 2). In their publication, Berman et al. (2008) reported that exposure to simulated natural settings restored executive attention as measured by the Attention Network Test in 12 young adults. In our article, we present a conceptual replication attempt with 31 young adults as well as meta-analytic evidence to show that simulated nature does not seem to reliably restore executive attention when combining evidence from 14 studies ($N = 612$) that also used the executive portion of the Attention Network Test. While simulated nature exposure may provide benefits for some cognitive tasks, our findings and those of other recent meta-analyses question the reliability of simulated nature settings to restore executive attention. We discuss suggestions for future research.

According to Kaplan's (1995) Attention Restoration Theory, directed attention can be restored by environments that provide *soft fascination* (i.e., the opportunity for effortless reflection), a feeling of *being away*, *extent* (i.e., expansive enough to occupy the mind), and *compatibility* with a person's purpose. Kaplan (1995) proposed that natural settings can embody these four features and can, therefore, restore directed attention. In an article published in *Psychological Science*, Berman et al. (2008; Experiment 2) reported that executive attention, a form of directed attention that can be measured using the Attention Network Test (ANT), was restored in participants who viewed virtual natural settings on a computer screen. However, recent meta-analyses (Bowler, Buyung-Ali, Knight, & Pullin, 2010; Ohly et al., 2016; Stevenson, Schilhab, & Bentsen, 2018) have provided little convincing evidence that nature can reliably restore directed or executive attention. Bowler et al.'s (2010) meta-analysis on the ability of real natural settings to restore attention found that attention performance was significantly higher after nature exposure compared to control settings. However, a follow-up meta-analysis revealed no significant improvement when controlling for pretest performance. Ohly et al.'s (2016) meta-analysis

on the ability of real and simulated natural settings to restore attention found only three of the 13 attention measures showed improvement following nature exposure. Hartig and Jahncke (2017) warned about over-generalizing these meta-analytic results given they were based on only a few studies, with many of the studies including small sample sizes. More recently, Stevenson et al. (2018) conducted several meta-analyses examining the influence of nature on a variety of cognitive tasks. Of most relevance to this study, Stevenson et al. combined results from seven attentional control (including executive attention) studies using three different cognitive tasks (Stroop, ANT, and Multi-Source Interference Task) and found no significant restoration following nature exposure when controlling for baseline performance. Stevenson et al. (p. 264) suggested the "need to synthesize a more homogenous set of studies to ensure more reliable synthesis." They also mentioned (p. 231) that the executive portion of the ANT is a "refined measure of directed attention that isolates the ability from other cognitive processes."

In the current article, we first report a representative attempt at a conceptual replication of Berman et al.'s (2008) Experiment 2 findings.

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Then we provide a meta-analysis of data collected from 14 other studies ($N = 612$) that used the executive portion of the ANT as the dependent measure of directed attention to test nature's ability to restore attention. No other meta-analysis has synthesized these data; therefore, it is novel.

1. Conceptual replication study

We sought to conceptually replicate Berman et al.'s (2008) Experiment 2 and hypothesized that simulated natural settings would restore executive attention better than urban settings. We made some methodological changes to Berman et al.'s (2008) Experiment 2 by testing a larger sample, offering a variety of natural settings so that participants could choose the one most restorative to them (i.e., improved *compatibility*), and including sounds with the presentation of pictures to provide a multi-modal nature experience. The study was not pre-registered because the authors' goal was to replicate the findings of Berman et al. to build upon and extend them in further studies.

1.1. Method

Participants. A convenience sample of 42 participants were recruited from three Psychology research methodology courses at DEIDENTIFIED University. Procedures used were in accordance with the Declaration of Helsinki and were approved by the local Institutional Review Board (DEIDENTIFIED University Number 2015-88). Participants were awarded extra credit for their participation. Of the 42 participants, 11 did not complete the protocol (four did not return Week 2; seven did not follow instructions) and were removed from the final analysis. Therefore, 31 participants (2 males, 29 females; $M_{age} = 20$, $SD_{age} = 1.4$, age range 19–27) completed the study. In an attempt to blind participants to the purpose of the study, they were told it was a test of their attention capacity and the nature restoration component was not mentioned. Researchers were not blind to the purpose of the study.

Materials and procedure. Participants were tested in a room containing five semi-private cubicles, each with a computer and a set of sound-attenuating headphones. Up to three participants were tested at a time between the hours of 9:00 a.m. and 3:00 p.m. Each participant was positioned in a manner so that they were not facing or distracting others. Participants wore the sound-attenuating headphones throughout the study to help block out extraneous noises. Participants also entered and exited the room as a group to keep distractions during testing to a minimum.

After providing informed consent Week 1, participants completed a 5-min written survey that included questions for gender, age, self-rated attention on a scale of 1 (*easily distracted*) to 7 (*rarely distracted*), self-reported diagnosis of ADHD, and amount of sleep the night previous. Participants then completed a shortened written version of the Positive and Negative Affect Scale (PANAS), in which they rated, on a scale of 1 (*not at all*) to 5 (*extremely*), the extent they felt sad, angry, happy, relaxed, tired, and alert at the current moment.

To determine each participant's preferred restorative setting, each was asked to rate color printed pictures from five different settings: beaches/oceans, forests/streams, mountains/lakes, farmland/rolling hills, and urban cities (see Supplemental Materials 1 for examples). Participants used a 5 min shortened version of the Perceived Restorativeness Scale (PRS; Berto, 2005) to rate the five settings, in which they rated *fascination*, *being away*, *extent*, and *compatibility* on scales of 0 (*not at all*) to 10 (*very much*) for each of the five settings. Then a total restorativeness score was calculated as the sum of those ratings (maximum score of 40).

After the PRS, participants completed a pretest ANT on desktop computers for about 20 min using Inquisit 4.0 by Millisecond software. Participants read the instructions, completed 10 practice trials, and then completed three runs of the ANT. Reaction times and accuracy were recorded.

After completing the pretest ANT during Week 1, participants were

randomly assigned to an environment intervention ($n = 17$ assigned to urban and $n = 14$ assigned to nature during Week 1). Twenty-five high quality, color photographs (640×480 resolution) chosen from Google images were presented in a PowerPoint slideshow that automatically transitioned every 15 s. The pictures were all the same size and filled the screen except for a small black border. A sound recording corresponding to the environment was presented through headphones at a level deemed comfortable to the participant (see Supplemental Materials 1 for sound recordings used). Beaches/oceans images included water, sand, palm trees, rock formations, sky, and a minimal number of human-made structures. Sounds included gently rolling waves. Forests/streams images included mostly green forest settings, some with autumn colors, half with streams, and some with forest paths. None included sky and none included human-made objects. Sounds were mostly birds chirping. Mountains/lakes images included mostly snow-capped mountains, some green-covered mountains, and sky. Half of the images contained water. A minimal number of the images included human-made objects, and the sound was breezes that sounded like a low frequency white noise. Farmland/rolling hills images included green grass, hay, or wildflowers, and included some human-made objects including barns, unpaved roads, and wooden fences. Sounds included birds chirping and gentle breezes. Urban city images included tall buildings, streets and traffic, people walking, and sky. Minimal amounts of nature included patches of grass, bushes, trees, and water. Urban sounds included car noises such as horn honking, braking as well as trucks, buses, and cars driving by. Some human sounds were also included such as footsteps, whistles, and indecipherable speech.

All participants randomly assigned to the urban intervention during Week 1 viewed the same 25 urban photographs and heard the same urban sounds. Those assigned to the nature intervention during Week 1 viewed the images they rated highest in restorativeness from the PRS scale. Each participant rated at least one nature setting as more restorative than urban cities. If an individual rated two nature environments equally high in restorativeness, the nature environment with the highest compatibility rating was presented. If the compatibility ratings were equivalent for the tied nature environments, the participant was asked which of the two nature environments they preferred to view. Overall, the pictures and sounds were presented for a total of 375 s (just over 6 min). Participants did not complete a task during the presentation. However, participants were provided the instruction, "Please envision yourself in the scene, and imagine exploring it." A researcher covertly monitored participants to confirm they were watching the images.

Immediately following the nature or urban environment intervention in Week 1, participants completed a written posttest PANAS for about 1 min, again rating the extent to which they felt sad, angry, happy, relaxed, tired, and alert at the current moment. Participants then completed the posttest ANT for about 20 min on the computer. Before leaving, participants were asked not to speak with other students about the study and were instructed to return in 1 week at the exact same time and location.

When returning Week 2, participants completed a short survey of self-rated attention and amount of sleep the night previous for 2 min. Participants then completed the same protocol as Week 1 except they experienced the opposite environment ($n = 14$ experienced urban and $n = 17$ experienced nature). Those completing the nature intervention in Week 2 were assigned the environment they rated highest in restorativeness on the PRS at Week 2 (i.e., not what they rated highest in Week 1). Overall, the 31 participants were assigned to the following nature conditions based on their PRS scores: $n = 12$ to oceans/beaches, $n = 8$ to mountains/streams, $n = 8$ to farmland/rolling hills, and $n = 3$ to forests/streams. After completing the Week 2 protocol, participants were individually debriefed and asked not to speak about the study with anyone until data collection was complete.

1.2. Results

Power analyses were run prior to conducting statistical analyses, ensuring adequate sample size. Self-reported attention on a scale of 1–7 ($M_{WEEK1} = 4.5$, $SD = 1.1$; $M_{WEEK2} = 4.6$, $SD = 1.0$) and sleep in hours ($M_{WEEK1} = 7.2$, $SD = 1.2$; $M_{WEEK2} = 7.1$, $SD = 1.0$) did not significantly differ from Week 1 to Week 2 (paired t -tests, $p > .05$). Two participants self-reported a diagnosis of ADHD; neither exhibited outlier performance on the executive portion of the ANT so both were included in the analysis.

Data from the shortened PRS (see Supplemental Materials 2), averaged over Weeks 1 and 2, showed that total restorativeness scores were higher for nature settings ($M_{BEACHES} = 31.2$, $SD = 6.2$; $M_{FORESTS} = 28.6$, $SD = 6.6$; $M_{MOUNTAINS} = 31.6$, $SD = 5.6$; $M_{FARMLAND} = 29.2$, $SD = 7.2$) than urban cities ($M_{URBAN} = 18.7$, $SD = 7.4$). Ratings of being away, extent, and compatibility were all lower for urban cities than nature settings. Interestingly, fascination ratings for urban cities were similar to fascination ratings for the nature settings.¹

Paired t -tests of pretest, posttest PANAS ratings (see Supplemental Materials 2) showed that relaxation significantly increased after the nature intervention ($M_{PRE} = 3.2$, $SD = 0.8$; $M_{POST} = 4.1$, $SD = 0.7$, $p < .001$) but relaxation significantly decreased after the urban intervention ($M_{PRE} = 3.3$, $SD = 0.7$; $M_{POST} = 2.6$, $SD = 0.8$, $p = .001$). Tiredness also significantly increased after the nature intervention ($M_{PRE} = 2.8$, $SD = 1.1$; $M_{POST} = 3.3$, $SD = 1.2$, $p = .008$). While tiredness increased after the urban intervention, it was not significant after correction for multiple comparisons ($M_{PRE} = 2.8$, $SD = 0.9$; $M_{POST} = 3.2$, $SD = 1.0$, ns); however, tiredness after the nature intervention did not significantly differ from tiredness after the urban intervention ($p = .60$). No other PANAS ratings demonstrated significant pretest, posttest differences using a corrected alpha level ($\alpha_{BONFERRONI} = 0.05/6 = 0.0083$).

The primary outcome measure for the ANT is reaction time (Fan et al., 2002).² Given our particular research interest, we only analyzed reaction time scores from the executive attention portion of the ANT (see Supplemental Materials 2 for means and standard deviations from executive, orienting, and alerting portions). Reaction times represent the cost of having to ignore incongruent stimuli around a central target (e.g., the 4 left-facing arrows must be ignored and attention must be paid to the central right-facing arrow $\leftarrow\leftarrow\leftarrow\leftarrow\rightarrow$ in incongruent trials) compared to a target surrounded by congruent stimuli (e.g., attending to the central arrow $\rightarrow\rightarrow\rightarrow\rightarrow\rightarrow$ is easier in congruent trials). A higher executive attention reaction time score represents worse performance in incongruent trials compared to congruent trials and suggests strain on the executive attention network.

We collected pretest and posttest ANT reaction times in milliseconds for correct trials in nature and urban conditions. We converted the data to difference scores (pretest minus posttest) so that a positive difference reflects an improvement at posttest (i.e., faster reaction to incongruent compared to congruent trials after the intervention). Fig. 1 shows the difference scores for nature and urban conditions as well as 95% confidence intervals. As seen in Fig. 1, the improvement in posttest reaction times was higher after the nature intervention ($M_{diffNATURE} = 7.4$ ms,

95% CI [18.8, −4.0]) than the urban intervention ($M_{diffURBAN} = 4.7$ ms, 95% CI [16.1, −6.7]) but the difference was not significant (1-tailed, paired t -test, $t(30) = 0.35$, $p = .36$).³

1.3. Discussion

The results showed that even though participants rated simulated nature settings as more restorative than simulated urban settings, the change in executive attention performance after the nature intervention did not significantly differ from the change in performance after the urban intervention. Our findings are, therefore, inconsistent with Berman et al.'s findings (2008, Experiment 2). It is possible that because Berman et al.'s participants completed the Backward Digit Span Task before completing the ANT, their participants had greater attentional fatigue and, therefore, a greater opportunity to show restoration effects after viewing pictures of nature. However, pretest ANT performance was similar in our study to Berman et al.'s suggesting that our participants were not noticeably less attentionally fatigued (Berman et al.'s nature group $M = 86$ ms ($SE = 11.3$) and our nature group $M = 84$ ms ($SE = 10$); Berman et al.'s urban group $M = 81$ ($SE = 15.5$) and our urban group $M = 81$ ms ($SE = 8.4$)).

While we replicated most of the methodology used by Berman et al. (2008, Experiment 2), we also included some changes. First, our sample size was 2.5 times greater than Berman et al.'s (2008, Experiment 2) and should have had the power to find a significant effect if one exists. Second, we personalized the nature setting for each individual in an attempt to provide the most compatible and restorative setting possible. No participant rated the urban setting as more restorative than their highest rated nature setting but some participants did rate the urban setting as more restorative than their least restorative nature setting. Therefore, we decided that providing one nature setting (e.g., forest scenes like Berman et al.) to all participants may not be as effective as providing each participant their most restorative nature setting. While our nature and urban pictures were presented for 15 s each versus 7 s each in the Berman et al. (2008) study, Berman et al. (2008) had participants rate how much they liked each picture after viewing it. Therefore, the total intervention time in their study may comport with the total intervention time in ours. However, it is possible that our longer presentation time may have induced fatigue and, therefore, negated some restorative benefits of the nature pictures. Because participants rated their tiredness after the nature and urban interventions, which directly followed completion of the pretest ANT, it is not possible to decipher whether the increased tiredness was due to the ANT or the total length of the intervention. Unlike Berman et al. (2008), we presented environmental sounds with the nature and urban pictures with the hope of making the interventions multi-modal and thereby stronger. It is possible that these differences between Berman et al.'s (2008) and our nature and urban interventions could account for the differences in our findings. Third, Berman et al. (2008, Experiment 2) administered the Backward Digit Span Task in addition to the ANT to assess attention restoration while we did not administer the Backwards Digit Span Task as this primarily measures working memory, according to factor analytic research (Lezak, Howieson, Loring, & Fischer, 2004). When utilizing complex cognitive tests, like the Backwards Digit Span Task, there is more variance introduced to the interpretation of the results that may not necessarily align with the conceptualization of the Attention Restoration Theory in comparison to 'purer' tests of attention, like the ANT. However, it is possible that because our participants did not complete the Backwards Digit Span Task before the ANT, they may have been less attentionally fatigued and, therefore, may have had less of an

¹ We further explored this in a separate sample of participants ($N = 34$) using the same set of pictures, and found that soft fascination (i.e., a place that provides opportunity for reflection) was higher for nature settings than urban settings, but hard fascination (i.e., a place large enough for me to discover and be curious about things) was similar for nature and urban settings. See Supplemental Materials 3.

² Accuracy data were not extracted from the original data files, and the original data files can no longer be accessed. It is, therefore, not possible to report participants' accuracy data.

³ Stenfors et al. (2019) found a stronger nature versus urban effect in the second session when analyzing Backward Digit Span Task data. We split our data by session as Stenfors et al. had done and failed to find a session difference. See Supplemental Materials 2.

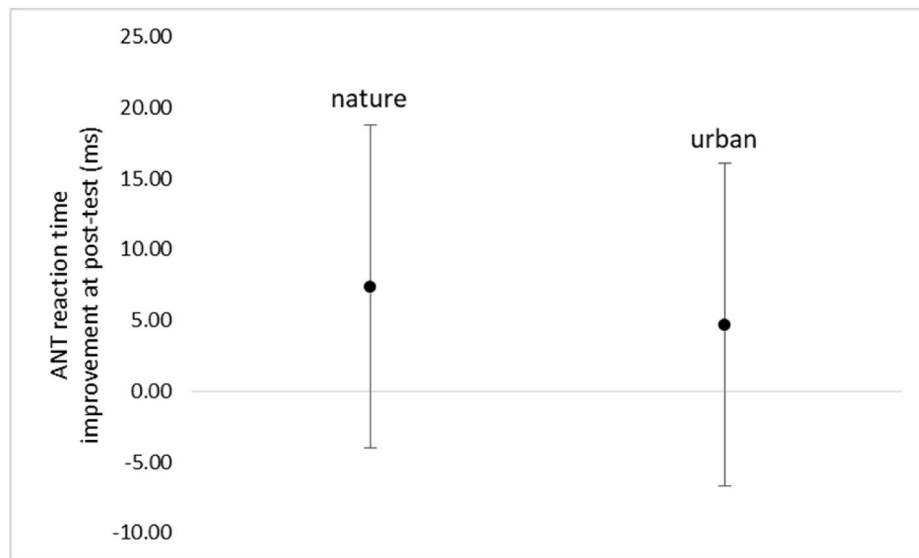


Fig. 1. Executive attention reaction time difference scores (pretest minus posttest) for nature and urban interventions (dots) and 95% confidence intervals (error bars) from the ANT are presented. Higher scores reflect an improvement at posttest (i.e., faster reaction time after intervention). The improvement was higher after the nature intervention than the urban intervention but the difference was not significant.

opportunity to show restoration effects. It may have been helpful to compare participants' accuracy scores from the ANT to Berman's study as speed can compromise the accuracy of responses (Posner, Klein, Summer, & Buggie, 1973); however, these data were not obtained in the current study. Lastly, we documented various details crucial to interpreting cognitive test performance that Berman et al. (2008, Experiment 2) did not. Examples of these details include documenting whether participants were excluded for having associated-attention deficit disorders, the proximity of the computer/screen to the participant, how many participants were tested at one time, time of day participants were tested (and ensuring that they were tested again at that same time), whether participants were blind to the purpose of the study to avoid placebo effects, measures of tiredness, age range of participants, and gender of participants.

2. Research synthesis and meta-analysis

We focused our synthesis and meta-analysis on results from studies that used the executive portion of the ANT to determine if nature could restore directed attention. Also, in an attempt to counter publication bias (i.e., tendency for non-statistically significant findings to be underrepresented in the published literature), we included Master's theses and Doctoral dissertations in our meta-analysis. Further, to include as many studies as possible in the meta-analysis, we included real and virtual nature interventions.

2.1. Method

We conducted the research synthesis and meta-analysis in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement (Moher, Liberati, Tetzlaff, Altman, & The PRISMA Group, 2009) as well as the American Psychological Association's Meta-Analysis Reporting Standards.

Eligibility criteria. To be eligible for inclusion in the meta-analysis, studies had to meet the following criteria:

- Population of interest: Physically and psychologically healthy female and male humans with normal or corrected-to-normal hearing and vision.
- Intervention: Contact with a natural setting, real or simulated. Simulated contact could be visual (e.g., pictures on a computer

screen) or aural (e.g., nature sounds heard through headphones) or both. [*Exclusion criterion:* There could be no concurrent activity during the intervention that has been shown to influence attention (i.e., meditation or exercise). We *did* include studies of people walking in nature or performing a Likert scale task during the intervention.]

- Comparator (control) group: Contact with an urban or built setting. The comparator group had to match the intervention in all other ways (i.e., real/simulated; visual/aural/both; length of contact; task performed during contact).
- Outcomes: Use of the executive portion of the ANT as the dependent measure. Pretest and posttest measures of performance were required.
- Study design: Random assignment to intervention and comparator group when an independent groups design was used; counterbalancing of intervention and comparator group when a repeated measures design was used.

Other criteria for eligibility included:

- Methods and results that were available in the English language.
- Studies published in journals, or studies reported in Master's theses or Doctoral dissertations. If data from a thesis were published in a journal article, the journal article version was used.
- Enough information provided to compute effect sizes (i.e., pretest and posttest reaction time means and standard deviations from the executive portion of the ANT).

Literature search. We conducted a preliminary literature search in February 2018 using Google Scholar, PsycArticles, PsycInfo, and ProQuest Dissertation & Theses. We used combinations of keywords in our searches: *attention network test/task*, *attention restoration*, *nature/natural setting*, *outdoor*, *urban*. The combination that led to the most successful result was *attention network t** and *attention restor** in Google Scholar. Reference sections of relevant reports were checked for additional sources. Also, forward searches of relevant studies were conducted in Google Scholar. In addition, we reviewed the reference sections from and did forward searches on three recent, relevant meta-analyses (Bowler et al., 2010; McMahon & Estes, 2015; Ohly et al., 2016). When we found studies that did not contain the information needed to conduct our meta-analysis, we requested that information from the authors. Nearly all authors responded and either provided the requested

information (Egner, 2016; Gamble, Howard, & Howard, 2014; Kuziek, 2017), or indicated that the data were no longer available (Andracchio, 2012; Rose, 2011). Only one author did not respond to our request. A final literature search was conducted in June 2018. No additional sources were found at that time.

Study selection and data collection process. The literature search yielded 783 potentially relevant records. After removing duplicates, review articles, conference abstracts, and records unavailable in the English language, 414 records remained (see Supplemental Materials 4 for flowchart of the search protocol and workflow). These records were screened by accessing the abstracts and/or method sections, and then searching for the inclusion criteria of the intervention (i.e., nature contact) and outcome (i.e., ANT used as dependent variable). This yielded 15 potentially relevant records (6 published articles and 9 theses). The second author conducted an independent literature search using the same databases and key words and found no additional relevant reports. The first and second authors then independently extracted the following information from the 15 potentially relevant records:

- Population of interest: Sample size; age; gender; recruitment procedure; attrition; knowledge of purpose of study.
- Intervention: real or simulated natural setting; type of natural setting; length of contact; task performed during contact.
- Comparator (control) group: real or simulated urban/built setting; type of urban/built setting; length of contact; task performed during contact.
- Outcomes: use of pretest/posttest ANT with intervention between; amount of time between intervention and posttest.
- Study design: Random assignment to intervention and comparator group when an independent groups design was used; counterbalancing of intervention and comparator group when a repeated measures design was used.
- Effect size information: Pretest and posttest reaction time means and standard deviations from the executive portion of the ANT.

After coding the 15 records, the two authors agreed that two published articles and four theses did not meet the inclusion criteria. Reasons for the exclusions are as follows. Haga et al.'s (2016) published article included intervention and comparison groups that received the same exact sound stimulus (i.e., pink noise); the intervention group was told the sound was from a natural setting and the comparison group was told the sound was from a built setting. The study was excluded because the intervention did not include an actual natural sound. Kelz et al.'s (2015) published article was excluded because it was a quasi-experimental design; participants were not randomly assigned to the intervention and comparator groups. Three theses were excluded because the information to compute effect sizes was not available (Andracchio, 2012; Rose, 2011; van der Jagt, 2014). Gomm's (2014) thesis was excluded because it did not include a pretest ANT measure.

Adequate agreement was reached for all coding variables for the remaining four published articles and five theses (Cohen's Kappa = .717 to 1.0; intraclass correlation coefficients = 0.996 to 1.0). Coding disagreements were discussed until agreed upon.

Of the nine remaining records, four included one intervention and control condition with one sample of participants; therefore, one effect size was calculated and included in the meta-analysis for each (i.e., Berman et al., 2008; Bratman, Daily, Levy, & Gross, 2015; Egner, 2016; Kuziek, 2017). The five other records included either multiple intervention conditions or multiple studies. To determine which intervention conditions and studies to compute effect sizes for and include in the meta-analysis, we followed this protocol.

- If a source included multiple studies that used independent samples, all studies were included in the meta-analysis. This applied to Gamble et al. (2014), who tested a young and an older sample of

participants, and Siu (2016), who tested two independent samples in Experiment 1 and Experiment 3.

- If a source included more than one nature intervention but only one control group, we included just one effect size from the source in order to maintain independence between effect sizes. We calculated effect sizes for both nature interventions separately and then chose the effect size that showed the strongest intervention effect (i.e., most attention restoration due to nature) to include in the meta-analysis. This applied to Michaelis (2013) and Larsen (2011). Larsen tested two nature interventions (green and water) and one control condition. We included the effect size for just the water intervention in the meta-analysis because it showed the stronger intervention effect. Michaelis (2013) also tested two nature interventions (foliage and aquatic) and one control condition. The foliage intervention was stronger and therefore included in the meta-analysis (Note. Michaelis (2013) also varied the color of the pictures; we only used the natural, untouched pictures in this analysis.)
- If a source included multiple treatment and control conditions that used independent samples, all were included in the meta-analysis as separate effect sizes. This applied to Emfield and Neider (2014). Three different nature interventions were each completed by separate groups of participants: one group viewed nature pictures, one group heard nature sounds, and one group viewed nature pictures and heard nature sounds. Each intervention condition had a matching control condition completed by separate groups of participants. Therefore, three independent effect sizes were calculated (one for pictures, one for sounds, one for both) and each was included in the meta-analysis.

In total 13 effect sizes from the literature plus an effect size computed from the conceptual replication study reported in this article were included in the meta-analysis (14 total effect sizes).

Summary measures and data analyses. We used Hedges' *g* as our effect size measure. Hedges' *g* is calculated and interpreted similarly to Cohen's *d* but is adjusted to account for small sample sizes using the formula $[1 - 3/(4 * (df - 1))]$. Hedges' *g* effect sizes were calculated using formulas for study designs that include a pretest and posttest as well as an intervention and control condition, that is, 2×2 designs (Morris, 2008). The general formula for calculating the effect sizes was $d = [(M_{\text{nature pretest}} - M_{\text{nature posttest}}) - (M_{\text{urban pretest}} - M_{\text{urban posttest}})] / SD_{\text{pooled}}$. We followed Morris' (2008) suggestion of using pooled pretest SDs in the denominator, which leads to a more precise estimate than other techniques. Effect sizes (*d*) were then adjusted resulting in Hedges' *g*. Effect sizes were calculated such that positive values indicated improved performance following the nature intervention compared to the urban control condition.

The standard error of each *g* was calculated using formulas suggested by Morris (2008) that were adjusted when the intervention and control conditions were repeated measures (see all formulas in Supplemental Materials 5). The 14 Hedges' *g* effect sizes and standard errors were combined using Meta-Essentials (Suurmond, van Rhee, & Hak, 2017) using a random effects model and inverse variance weighting. Heterogeneity of effect sizes was assessed using I^2 , and publication bias was assessed using funnel plot and trim-and-fill analyses.

2.3. Results

Qualitative summary of studies included in the meta-analysis.

Fourteen independent effect sizes from 10 sources (four published articles, five theses, and the conceptual replication study reported in this article) were included in the meta-analysis (see Supplemental Materials 6 for summaries). One study used a real natural intervention for 50 min (Bratman et al., 2015); all other studies included simulated natural settings presented in a laboratory for between 6 and 12 min. All but one simulated natural setting involved the presentation of visual nature

images on a computer screen; two studies included natural sounds presented simultaneously through headphones with the visual images (Emfield & Neider, 2014 and the conceptual replication study reported in this article); one included just natural sounds through headphones without simultaneous visual images (Emfield & Neider, 2014). Three sources (four effect sizes) used the same visual images as Berman et al. (2008). Most simulated nature interventions were compared to a control condition that included urban scenes of buildings, streets, and traffic. During most nature interventions, participants were asked to complete a task such as rating how much they liked each image or how relaxing each image was. Only Larsen (2011) and the conceptual replication study reported in this article did not require participants to complete a task during the nature intervention (i.e., participants were only instructed to imagine themselves in the scenes).

In all studies, participants complete a pretest ANT before the intervention and a posttest after. In most cases, the posttest occurred within a few minutes of completing the intervention. Exceptions included Bratman et al. (2015) and Emfield and Neider (2014). Most researchers used the full 288 trial version of the ANT for both the pretests and posttests. Exceptions included Emfield and Neider (2014) and Kuziek (2017) who used shortened versions. Gamble et al. (2014) had compared the third run of the pretest to the first run of the posttest in their data analysis, but they provided us with the data for all three runs, which we combined and used in the calculation of effect size for this meta-analysis in order to be consistent with the other studies included in the meta-analysis.

On average, mixed design studies ($k = 9$; intervention as a between factor; pretest/posttest as a within factor) included 42 participants ($SD = 16$) and repeated designs ($k = 5$; intervention and pretest/posttest as within factors) included 23 participants ($SD = 8$). The average age of participants in all but one study (Gamble et al., 2014, $M_{age} = 69$) ranged from 19 to 24 years old ($M = 21$, $SD = 2$). On average, studies included 67% females ($SD = 12$). Among the 10 sources, six claimed not to find support for the Attention Restoration Theory and four claimed to find support for it.

Tests of heterogeneity. Effect sizes ($k = 14$, # of observations = 612) ranged from Hedges' $g = -0.22$ to Hedges' $g = 1.10$, and were substantially heterogeneous, $I^2 = 63\%$ (see Fig. 2). This justified testing for moderator effects. As per Morris' (2008) suggestion, we first tested the influence of design on the outcome by comparing 2×2 mixed designs ($k = 9$) to 2×2 repeated designs ($k = 5$). Mixed designs showed considerable homogeneity, $I^2 = 0\%$, demonstrating that the effect sizes could be reasonably combined in a meta-analysis. Repeated measures designs were heterogeneous, $I^2 = 81.4\%$, suggesting combining effect sizes in a meta-analysis was not warranted. Due to the low number of repeated measures designs, further tests of moderator effects were

unreasonable to explore.

Effect of nature on attention restoration in mixed designs. Because the effect sizes from mixed designs were homogeneous, they were combined ($k = 9$, # of observations = 382). The combined effect size estimate indicated no improvement in attention following the nature intervention compared to the control condition (Hedges' $g = -0.04$, $SE_g = 0.07$, 95% CI $[-0.20, 0.13]$). As seen in Fig. 2, the 95% confidence interval of the combined effect size overlaps with 0 indicating that there was no effect of the nature intervention. Publication bias was investigated using a funnel plot and trim-and fill analysis, both of which indicated that no effect sizes were missing from our analysis (i.e., there was no publication bias).

Effect of nature on attention restoration in repeated measures designs. Given the high degree of heterogeneity among effect sizes from repeated measures designs ($k = 5$, # of observations = 230, $I^2 = 81.4\%$), the combined effect size estimate should be interpreted cautiously (Hedges' $g = 0.54$, $SE_g = 0.19$, 95% CI $[0.02, 1.07]$). Suurmond et al. (2017) suggested that when heterogeneity is high, an adjusted confidence interval of the effect size should be used known as the prediction interval (95% PI $[-0.83, 1.92]$). The 95% prediction interval has considerable overlap with 0 indicating no improvement in attention following the nature intervention compared to the control condition. A test of publication bias was not conducted as it is not appropriate to do in a set of heterogeneous effect sizes.

Influence of publication status on outcome. One could argue that studies published in journal articles have undergone a strict peer-review process and could therefore be of higher quality than unpublished work. To determine if publication status influenced the outcome of the meta-analysis, we used it as a moderator variable. Effect sizes from published works ($k = 7$, # of observations = 300) were homogenous ($I^2 = 0\%$). The combined effect size estimate indicated no improvement in attention following the nature intervention compared to the control condition (Hedges' $g = 0.08$, $SE_g = 0.11$, 95% CI $[-0.19, 0.35]$). Effect sizes from unpublished works ($k = 7$, # of observations = 312) were heterogeneous ($I^2 = 79\%$); therefore, combining effect sizes proceeded cautiously. The effect size estimate indicated no improvement in attention following the nature intervention because the prediction interval overlapped with 0 (Hedges' $g = 0.31$, $SE_g = 0.18$, 95% CI $[-0.14, 0.76]$; 95% PI $[-0.84, 1.46]$).

3. Conclusions

Attention Restoration Theory (Kaplan, 1995) suggests directed attention can be restored by environments with *soft fascination*, a feeling of *being away*, *extent*, and *compatibility*. Accordingly, natural settings that

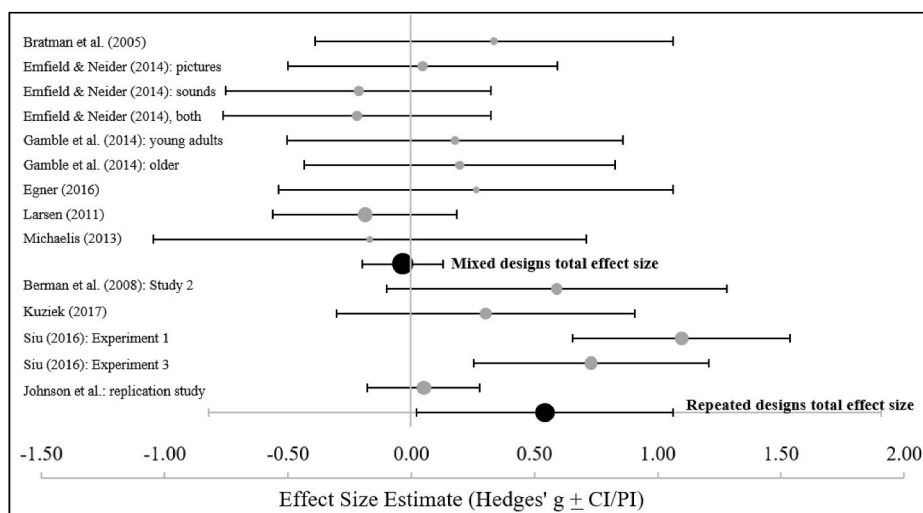


Fig. 2. The gray circles are the 14 independent effect sizes and the size of each circle is proportionate to the number of observations included in the estimate. Black circles are the total combined effect sizes for mixed design studies (homogeneous) and repeated measures design studies (heterogeneous). The black error bars are 95% confidence intervals. The gray error bars on the repeated designs total effect size are 95% prediction intervals. The combined effect size estimate for mixed design studies overlaps with 0 suggesting no restoration in attention after a nature intervention. The combined effect size estimate for repeated measures studies was heterogeneous suggesting the wider prediction interval, which overlaps with 0, is a more cautious estimate of the effect.

embody these features have the potential to restore directed attention. Previous meta-analyses (Bowler et al., 2010; Ohly et al., 2016; Stevenson et al., 2018) provided little evidence that natural settings reliably restore directed attention. Our conceptual replication study and meta-analysis add to this conclusion by showing, through synthesis of 10 years of research conducted by 10 independent research groups, that simulated nature does not reliably restore executive attention.

We are not suggesting nature bestows no benefits. In our conceptual replication study, simulated nature increased self-reported relaxation, and in two recent meta-analyses, self-reported emotions improved after nature exposure (Bowler et al., 2010; McMahon & Estes, 2015). Notably, McMahon and Estes (2015) found improved positive affect was greater with exposure to real versus simulated nature. Neither are we suggesting that the Attention Restoration Theory is falsified by our findings. Real nature was used in only one study in our meta-analysis, therefore, future researchers should explore whether real nature settings may more reliably restore executive attention than simulated ones. Furthermore, Stevenson et al. (2018) and Stenfors et al. (2019) found significant improvements in working memory tasks with high attention demands after nature exposure. Yet given our meta-analytic findings from the executive portion of the ANT, a “refined measure of directed attention that isolates the ability from other cognitive processes” according to Stevenson et al. (2018, p. 231), we think it is reasonable to conclude that simulated nature does not reliably restore directed attention.

Many questions still remain. Future researchers could consider exploring if there is a certain length of nature exposure required for reliable attention restoration. Perhaps there is an optimal exposure time that varies by personal preference, task demands, and demographic differences. In addition, future researchers must carefully equate nature and control interventions. Larsen (2011) noted differences in quality, decade, sunlight, brightness, and contrast between Berman et al.’s (2008) nature and urban photographs, which were used in five of the 14 effect sizes we reported. Such differences can confound results. Other factors to consider when choosing nature and control stimuli include people’s personal preferences as well as the presence of water, human-made objects, and sky.

We believe researchers should continue to heed the warnings of Hartig and Jahncke (2017) about over-generalizing meta-analytic results. While our mixed design effect size estimates were homogeneous, the considerable heterogeneity of the repeated design effect size estimates suggests further exploration is needed. Notably, whether participants were blind to the study’s purpose was mentioned by none of the repeated design authors. Given that participants completed both the nature and control interventions, it is possible that in some cases, participants guessed the purpose of the study. We encourage researchers to continue to use pretest/posttest designs to examine changes in attention performance, test a comprehensive battery of attention measures, and consider exploring how one’s level of attentional fatigue influences attention restoration (Hartig & Jahncke, 2017). With many questions remaining, it is important that we continue exploring this topic.

Author contributions

J. A. Johnson, E. L. Funk, F. L. Elezovic, and J. C. A. Finley developed the conceptual replication study design. Conceptual replication study data were collected and analyzed by E. L. Funk, F. L. Elezovic, and J. C. A. Finley and were interpreted by those authors in addition to J. A. Johnson. J. A. Johnson and B. Hansen reviewed and coded the studies used in the meta-analysis. J. A. Johnson analyzed the meta-analytic data and drafted the manuscript. All authors provided critical revisions and approved the final version of the manuscript for submission.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvp.2021.101709>.

Declaration of competing interest

The authors declared that they had no conflicts of interest with respect to their authorship or the publication of this article.

Open practices

Neither the conceptual replication study nor the meta-analysis reported in this article were formally preregistered. Neither the data nor the materials have been made available on a permanent third-party archive. Requests for data or materials can be sent via email to the lead author at DEIDENTIFIED.

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